

Package #162

Apply a heparin■coated polymer layer to the catheter lumen and integrate a thin piezoelectric strip that emits low■frequency ultrasonic waves (acoustic streaming) along the inner wall to synergistically inhibit clot adhesion.

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'catheter flow occlusion' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency within weeks of insertion despite flushing protocols?". [fda_maude] 100 records retrieved live (1992-2032). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 100000: Injury [VITAL SIGNS]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2026-2026). Illustrative records: 'Commentary on: "clinical and haemodynamic evaluation of a novel physician modified inner branch iliac branch device for East Asians".'; 'Biosurfactants as Antibiofilm Agents for Medical Devices: Mechanisms, Evidence and Integration into Infection Prevention and Control.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Anticoagulant heparin coating on the catheter surface reduces thrombus formation, leading to longer failure■free survival.

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.